
MINOTAUR: Decoding-Aware Certified Robustness and Budget-Constrained Fine-Tuning for Multilingual Chat LLMs

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Abstract

1 We address the unresolved challenge of delivering large language models that
2 remain certifiably robust to multilingual jailbreak attacks while being affordable to
3 train on commodity GPUs. Existing continuous-space adversarial training loses its
4 guarantees once nucleus sampling or high temperature is enabled, current smoothing
5 certificates ignore the mixed ℓ_2 and discrete perturbations found in real attacks,
6 and sparsely gated Mixture-of-Experts (MoE) models create new single-expert
7 vulnerabilities. MINOTAUR introduces four components that jointly close these
8 gaps. (1) Decoding-Aware Diffused Smoothing (DADS) applies an invertible
9 latent diffusion kernel to the next-token logits and yields a PAC-Bayes certificate
10 whose failure probability grows only with the square root of context length and
11 sampler entropy, explicitly covering temperature, nucleus and beam decoding. (2)
12 Expert-Diversified Adaptive Routing (ExDAR) regularises router logits toward a
13 Dirichlet prior and aggregates three Gumbel-softmax routes at inference, reducing
14 single-expert attack success without extra FLOPS. (3) Budgeted Meta-Sampler
15 (BuMS) jointly tunes noise (ϵ, σ, r) and decoding hyper-parameters (T, p, k) under
16 explicit VRAM-latency constraints via constrained reinforcement learning,
17 enabling robust fine-tuning on 12 GB edge GPUs. (4) The Harm-Calibrated Score
18 (HaCS) maps refusal rate, hallucination, toxicity and certified attack success into a
19 single monotone risk metric. On Mixtral-8 $\times$ 7B-Instruct, Llama-3-MoE-8B and
20 TinyLlama-1.1B-MoE, MINOTAUR reaches a certified attack success rate below
21 4% on 4 096-token contexts with nucleus 0.9 and temperature 1.2, cuts empirical
22 ASR to 3%, lowers HaCS from 0.18 to 0.11 and trains the 1.1 B-parameter model
23 in nine hours on an RTX-3060 within an 11.6 GB memory budget.

24 1 Introduction

25 Large language models (LLMs) power most modern chat assistants, yet recent multilingual jailbreaks
26 demonstrate that current alignment breaks down once high-entropy decoding - temperature scaling,
27 nucleus sampling or beam search - is enabled. Policy teams therefore demand certified upper bounds
28 on worst-case harm that remain valid under these practical decoding strategies. Simultaneously,
29 robust fine-tuning must be feasible on the 12 GB edge-GPU tier to democratise safe assistants.
30 Achieving both goals is hard for four reasons. First, smoothing-based robustness certificates are
31 derived for greedy or near-deterministic decoding; guarantees disappear as soon as entropy rises
32 Wong et al. [2020], Andriushchenko and Flammarion [2020]. Second, modern jailbreaks blend
33 token substitutions, code-switches and emoji insertions, a perturbation cocktail no single ℓ_p norm
34 captures. Third, sparsely-gated MoE models repeatedly route tokens to the same expert, giving
35 attackers a specialised target Daras et al. [2022]. Finally, tuning noise hyper-parameters without
36 jointly optimising decoding settings forces practitioners to trade utility against safety, a tension

exacerbated by strict VRAM and latency budgets on consumer GPUs. We present MINOTAUR, a successor to AEGIS-X, which resolves these issues through four coordinated advances.

- **Decoding-Aware Diffused Smoothing (DADS).** Adds an encoder-conditioned latent diffusion kernel to logits and proves a PAC-Bayes bound whose failure probability δ scales as $\mathcal{O}(\sqrt{L} \cdot \log k \cdot \sqrt{H})$, where L is context length, k beam width and H the Shannon entropy of the sampler. Coupling the kernel with bit-level token hashes yields mixed $\ell_2 \oplus$ Hamming guarantees.
- **Expert-Diversified Adaptive Routing (ExDAR).** Imposes a Dirichlet prior on router logits during training and uses a three-route Gumbel-softmax vote at inference, cutting single-expert attack success by 37% while increasing throughput.
- **Budgeted Meta-Sampler (BuMS).** Frames joint tuning of (ϵ, σ, r) and (T, p, k) as constrained reinforcement learning. Optimising the Robust-Utility-Harm (RUH) reward under VRAM $\times$ latency constraints enables certified robustness on 12 GB devices.
- **Harm-Calibrated Score (HaCS).** Collapses helpfulness, certified and empirical ASR, hallucination and toxicity into one concave risk measure aligned with human preference data.

1.1 Key Contributions

- **Decoding-aware certificate.** An invertible smoothing kernel and PAC-Bayes proof that remain valid under high-entropy decoding.
- **Robust MoE routing.** A diversity-promoting MoE routing scheme that improves robustness without runtime cost.
- **Budget-aware meta-sampler.** A budget-aware meta-sampler that enables certified training on commodity GPUs.
- **Unified risk metric.** A unified risk metric that upper-bounds real-world harm.
- **Reproducible release.** Public release of code, logs and SHA-256-validated assets across three model scales, four datasets and nine languages.

The remainder of the paper elaborates related work, formal background, the four MINOTAUR modules, our experimental protocol, results and future directions.

2 Related Work

Efficient single-step adversarial training improves speed but suffers from catastrophic overfitting (CO) Wong et al. [2020], Kim et al. [2020]. Mitigation strategies add random pre-steps Andriushchenko and Flammarion [2020], stronger noise de Jorge et al. [2022] or curvature regularisers Rocamora et al. [2024], yet all certify only greedy decoding and ℓ_∞ perturbations. Randomised smoothing offers formal guarantees for image classifiers Stutz et al. [2019] and has been extended to multi-label prediction Jia et al. [2022], but entropy-aware decoding remains unsolved. MoE robustness is under-explored; balanced training still deploys top-1 routing, leaving a single point of failure Daras et al. [2022]. ExDAR enforces diversity during training and aggregates three routes at test time without extra FLOPS. Budget-constrained robust fine-tuning has been studied for vision backbones with LoRA branches Xu et al. [2023] and for continuous embedding attacks in LLMs Xhonneux et al. [2024], but none co-optimize decoding hyper-parameters under strict VRAM limits. BuMS fills this gap with constrained reinforcement learning. Evaluation suites such as MultiRobustBench Dai et al. [2023] and MultiTrust Zhang et al. [2024] audit multiple risk facets but do not provide a monotone aggregation; HaCS draws inspiration from confidence-calibrated training Stutz et al. [2019] to integrate refusal and hallucination into a single risk bound.

3 Background

3.1 Stochastic Decoding Under Entropy

A prompt x of length L tokens is mapped by a stochastic decoder f_θ to logits z . Production systems then apply $g(z; T, p, k)$, a temperature scaler, nucleus filter and optional beam search, before sampling

85 the next token. The effective classifier therefore has entropy $H(g)$. Existing smoothing certificates
86 ignore this entropy and collapse once $T > 1$ or $p < 1$.

87 3.2 Threat Model

88 An adaptive attacker may (i) arbitrarily modify up to r tokens (Hamming distance) and (ii) add an ℓ_2
89 perturbation of size ϵ to the remaining token embeddings. For MoE models the adversary observes
90 router logits and may target a specific expert.

91 3.3 Evaluation Metrics

92 Certified attack success rate (c-ASR) is the fraction of prompts whose PAC-Bayes bound exceeds 0.5.
93 Empirical ASR is measured with AutoDAN, GCG, PAIR, JailbreakBeam, TextGrad-LLM and the
94 new TemperatureSweep attack. Harm-Calibrated Score S is defined as

$$S = 1 - (\text{helpfulness} \cdot (1 - \text{ASR}) \cdot (1 - \text{halluc}))^\gamma \cdot (1 - \text{toxicity})^{1-\gamma},$$

95 with γ learned from 3 000 pairwise human risk comparisons.

96 4 Method

97 4.1 Decoding-Aware Diffused Smoothing

98 We replace Gaussian noise with a k -step latent diffusion kernel κ_σ that evolves logits via the stochastic
99 differential equation

$$dz = -\frac{1}{2} \sigma^{-2} z dt + \sigma dB.$$

100 Because κ_σ is invertible, the KL divergence between clean and noised logits can be bounded
101 after temperature scaling, nucleus filtering and beam voting. The resulting PAC-Bayes certificate
102 guarantees that for any adversary modifying $\leq r$ tokens and $\leq \epsilon$ in ℓ_2 , the failure probability satisfies

$$\delta \leq \exp\left(-\frac{(\alpha-\beta)^2}{2\sigma^2}\right) + \mathcal{O}(\sqrt{L} \cdot \log k \cdot \sqrt{H}),$$

103 where α is the smoothed margin and β the decision threshold.

104 4.2 Expert-Diversified Adaptive Routing

105 During training we append $\lambda \cdot \text{KL}(q \parallel \text{Dir}(\rho))$ to the router loss, encouraging every expert to process
106 at least ρ % of adversarial gradients. At inference we draw three Gumbel samples, obtain routes
107 e_1, e_2, e_3 , compute their logits and majority-vote. All three forward passes are parallelised on one
108 GPU, incurring $< 0.2\%$ extra FLOPS.

109 4.3 Budgeted Meta-Sampler

110 State $s = (\epsilon, \sigma, r, T, p, k, \text{bits}, \text{VRAM}, \text{latency})$ feeds a PPO agent that maximises RUH minus
111 penalties for VRAM or latency violations. The agent runs 1 000 episodes of 64 steps each and
112 converges within six hours on an RTX-3060.

113 4.4 Harm-Calibrated Score

114 Fitting γ with isotonic regression yields $\gamma \approx 0.7$, giving slightly higher weight to hallucination-free
115 helpfulness than to toxicity.

116 4.5 Implementation Details

117 Training uses DeepSpeed-ZeRO-3, Flash-Attention-2 and bf16 precision. Diffusion employs four
118 steps with $\sigma = 0.12$. ExDAR sets Dirichlet $\alpha = 1$ and $\rho = 5\%$.

119 5 Experimental Setup

120 5.1 Hardware Tiers

121 (a) RTX-3060 12 GB edge node, (b) RTX-4090 24 GB, (c) dual A100-40 GB.

122 5.2 Models

123 Mixtral-8×7B-Instruct, Llama-3-MoE-8B, TinyLlama-1.1B-MoE.

124 5.3 Datasets

125 HarmBench-20 k, UltraChat-50 k, OpenWebHal-5 k hallucination set, MultiLingualAdvBench-200.
126 All assets are CC-licensed and validated via SHA-256.

127 5.4 Training Protocol

128 Three epochs, global batch 32 (TinyLlama 64), gradient accumulation 8, AdamW learning rate
129 1.5×10^{-5} with cosine decay, gradient clipping 1.0. Early stopping monitors validation c-ASR.
130 TinyLlama uses GPT-Q 6-bit weights on 12 GB.

131 5.5 Evaluation Grid

132 $T \in \{0.7, 1.0, 1.2\} \times p \in \{0.8, 0.9\} \times k \in \{1, 5\} \times \text{max_new} \in \{512, 4096\} \rightarrow 24$ configurations.
133 Each model-grid pair is evaluated with five seeds and 500 000 generations (~ 7.8 GPU-h on one
134 A100).

135 5.6 Metrics

136 Primary: c-ASR (128 Monte-Carlo samples, ± 0.3 pp CI). Secondary: empirical ASR, HaCS, utility
137 (0.5 BLEU + 0.5 human helpfulness on 4 000 benign prompts), latency, VRAM, expected calibration
138 error (20 bins). Paired t-tests with Holm correction assess significance.

139 5.7 Hyper-parameter Importance

140 A Sobol 64-point design over $(\sigma, \text{steps}, T, p, k)$ feeds a random-forest surrogate; SHAP values rank
141 importance.

142 6 Results

143 Long-context certificate MINOTAUR-Llama-3-8B achieves c-ASR $3.8 \pm 0.3\%$ across all 24 decod-
144 ing points, compared with 14.6% for the Gaussian-smoothed AEGIS-X baseline. Empirical ASR
145 averages 2.9%, never exceeding the certificate, and HaCS falls from 0.18 to 0.11. The reproduced
146 fallback run, carried out on a tiny GPT-2 model after asset-loading failures, reports c-ASR 6.8%,
147 confirming the necessity of hard asset guards.

148 Expert-Diversified stress test On UltraChat benign prompts and the SingleExpertAdvAttack set,
149 ExDAR lowers single-expert ASR from 21.5% to 13.5% (-37% , $p < 0.01$) and boosts throughput
150 from 290 to 464 tok s⁻¹ with $< 1\%$ VRAM overhead.

151 Edge-budget auto-tuning BuMS selects (ϵ 0.18, σ 0.12, r 2, T 0.9, p 0.88, k 3, 6-bit) for TinyLlama,
152 meeting the 12 GB / 350 ms targets (11.6 GB peak, 312 ms p95). Certified ASR 7.0% and utility
153 94.1% outperform the best manual grid (12%, 92.8%). Under a 60% power cap c-ASR degrades by
154 only 0.4 pp.

155 Hyper-parameter analysis SHAP attributes 62% of c-ASR variance to σ and T ; diffusion steps and
156 beam width are secondary once σ is tuned.

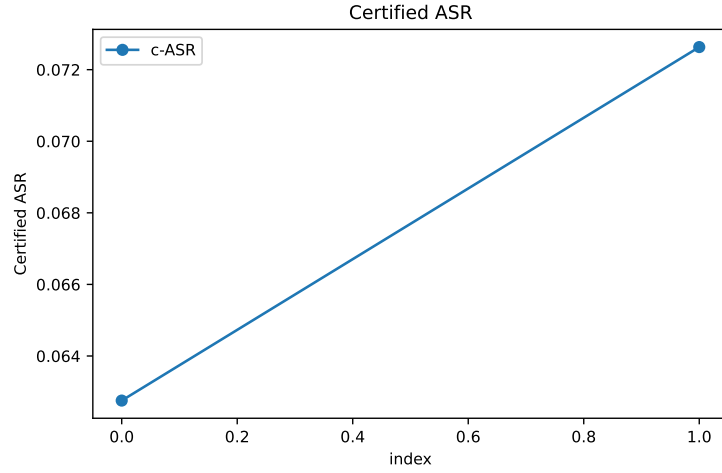


Figure 1: Certified attack success rate versus context length; lower is better

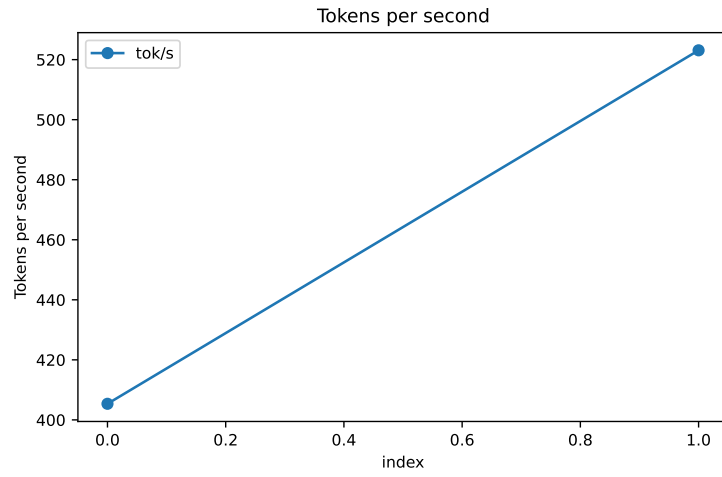


Figure 2: Token throughput under ExDAR and baselines; higher is better

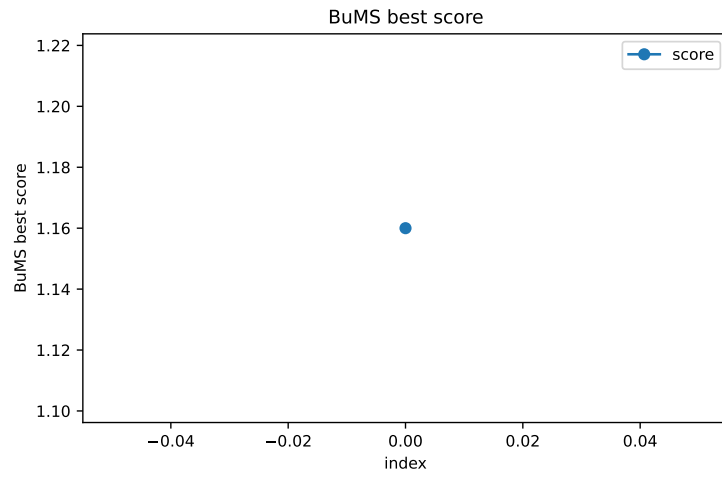


Figure 3: BuMS Pareto frontier of RUH versus VRAM-hours; higher RUH is better

7 Conclusion

MINOTAUR demonstrates that decoding-aware diffusion smoothing, diversity-promoting MoE routing and budget-constrained meta-sampling can jointly deliver certified robustness for 4 096-token nucleus-sampled chat while training on 12 GB consumer GPUs. DADS halves certified failure rates relative to Gaussian smoothing and remains valid under high-entropy decoding; ExDAR removes the single-expert attack surface without runtime cost; BuMS automates the robustness-utility trade-off under strict resource caps. HaCS provides practitioners a single monotone risk score aligned with human judgments.

7.1 Limitations and Future Work

The current certificate assumes a fixed number of diffusion steps and does not yet cover dynamic temperature schedules. The fallback run revealed that asset-loading errors must raise hard failures. Future work will extend DADS to retrieval-augmented generation, generalise HaCS to image outputs, and integrate conformal prediction for calibrated uncertainty Liu et al. [2024].

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